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Classification And Segmentation Of Brain Tumor Using PNN and CNN

1st M.Suneetha
Assistant Professor, Department of ECE
GRIET
Hyderabad, Telangana, India
suneetha645@gmail.com

2nd Bhupathi Raju Sai Likitha
Department of ECE
GRIET
Hyderabad, Telangana, India
likithavarma3455@gmail.com

3nd Sannapureddy Sai Spandana
Department of ECE
GRIET
Hyderabad, Telangana, India
spandana932@gmail.com

4th K.Madhu
Department of ECE
GRIET
Hyderabad, Telangana, India
Madhu.kandukuri683@gmail.com

5th Chindam Sharath
Department of ECE
GRIET
Hyderabad, Telangana, India
sharathchindam57@gmail.com

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***Abstract*—Automated fault identification in medical imaging has become a major issue in a range of biomedical applications. MRI tumor identification is important because it provides information about abnormal tissues, which is crucial for therapy planning. The most common method for finding abnormalities in magnetic resonance brain imaging is human examination. This method is impractical because of the massive amount of data. As a result, in order to reduce human mortality, reliable and automatic classification techniques are required. By using this tumor Identification methods are being developed to save radiologists time while maintaining tested accuracy. MRI brain tumor identification is complex due to the intricacy and variety of tumors .Deep learning methods are suggested in this study to address the disadvantages of existing classifiers in the detection of tumors in brain MRI using deep learning techniques. MRI may be used to detect cancer cells in the brain using deep learning and image classifiers.**

***Index Terms*—Preprocessing, DT-CWT, GLCM, Convolutional Neural Network, Probabilistic Neural Network, SFCM**

I. Introduction

The human brain contains myriad number of cells. The additional mass of cells turns into a tumor when cell expansion becomes unchecked. CT and MRI scans are utilized to determine the tumor's location.The main aim of our project is to detect the brain tumor accurately and to classify the tumor based on its region and area. Neurosurgeons, radiologists, and other healthcare professionals can use this technology. Using the industry standard simulation software tool MATLAB, the system is projected to improve the sensitivity, specificity, and diagnostic efficiency of brain tumour screening. These techniques involve pre-processing of MRI scans collected from online cancer imaging archives as well as scans obtained from several pathology labs. Images are resized and then we apply the proposed algorithms for segmentation and classification. The system is expected to improve the brain tumor screening procedure currently at use, and possibly reduce health care costs by decreasing the need for follow-up procedures. For accurate identification and analysis of medical imaging data, several processing steps are required.

II. Existing System

Sungheetha, Akey, and Rajesh Sharma achieved the research work which detects the diabetics of patients by extracting features from retinopathy images through Convoltional neural network framework [1].Morphological scanning, clustering, and Naive bayes classification are used to identify brain tumor in this technique [6].

Manoharan and Samuel are developed the automated system for detect the pulmonary nodules to find the cancer stage [2].In this paper axial slices method extracts features and they are fed to recurrent neural networks [7].

Dr Lenin Fred, A. Daniel and Ashy research work compares SFCM and K means clustering segmentation techniques to find the efficient and noise removal method to detect tumor in MRI [3]. By using superpixel zoning for segmentation in this project tumor region is identified automatically [8].

Saxena P, Maheshwari A and Maheshwari S research provides predicctive model to classify the brain tumor in MRI/CT scans. In this reseach paper, pretrained models such as Resnet, VGG-16, Inception-V3 are compared based on accuracy [4]. In this research work MRI brain images are taken as input anisotropic filtering is used for noise removal, SVM is used for segmentation and tumor region is seperated by using morphological operation. [9].

The research work provided by Busa S, Vangala N.S, Grandhe P and Balaji V. explains the simple algorithm for detect and segmentation of a tumor by using Fuzzy clustering and SVM [5]. This paper work is based on K-means and ANN techniques which identifies tumor part of the brain with the help of MATLAB [10].

III. Abbreviations

CNN - Convolutional Neural Network

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PNN - Probabilistic Neural Network
DWT - Discrete Wavelet Transform
DT-CWT - Dual Tree Complex Wavelet Transform
GLCM - Gray Level Co-Occurrence Matrix
SFCM - Spatial Fuzzy C-Means

IV. DATASET

The dataset which contains MRI brain images are taken from kaggle website. This dataset contains a Brain MRI scans . The Dataset is classified into training and testing datasets. The data set is divided into two categories: Glioma, Meningioma, Notumor, Pituitary. Different MRI pictures of the patients can be found in folders. Patients with brain tumours are in Glioma, Meningioma and Pituitary folder, whereas patients without brain tumours are in the Notumor folder. There are a total of 900 images of brain tumour positive patients and 300 images of patients who do not have a brain tumour. The images are all 256X256 pixels in size.

V. BLOCK DIAGRAM

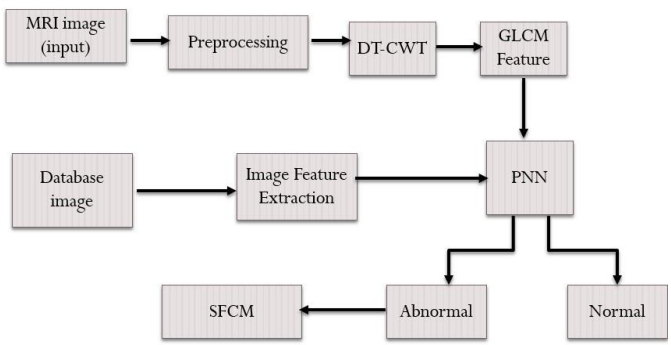

Fig. 1. Block diagram of Proposed system with PNN.

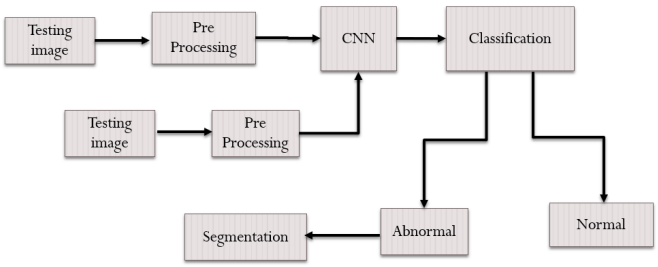

Fig. 2. Block diagram of Proposed system with CNN.

VI. PROPOSED SYSTEM

A. Preprocessing

Binary Thresholding, picture rescaling, noise removal, and morphological operations like dilation and erosion, as well as contour forming, are all part of our pre-processing. In the first step of pre-processing, memory space of image is decreased by reducing the gray-level of the pixels in the ranges 0-255. Because the shape of the brain is not segmented as tumors in this example, we employed the Median filter in noise removal because it is known to yield better results than the Gaussian filter.

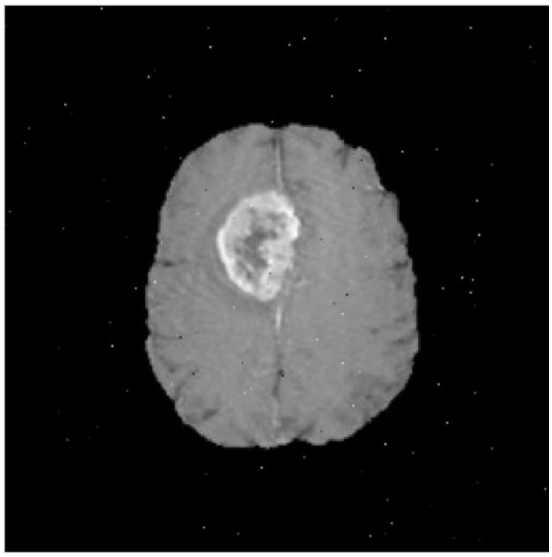

Fig. 3. Before Pre-processing with Noise

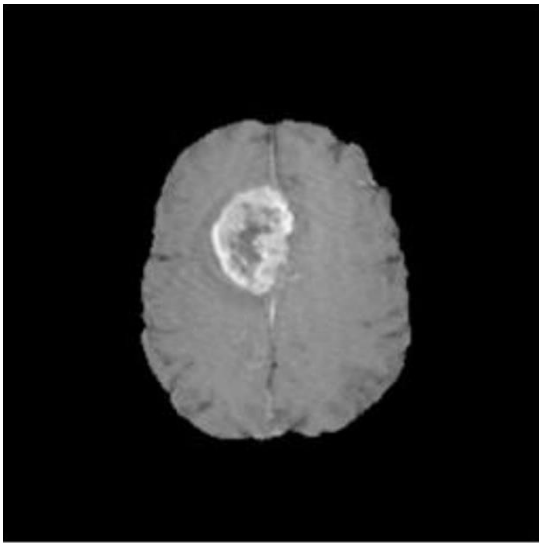

Fig. 4. After Pre-processing without Noise

B. Dual-tree Complex Wavelet Transform(DT-CWT)

Discrete wavelet transform is used to extracts features from brain tumor. DT-CWT is implemented as two separate two-channel filter banks. It is an enhancement to DWT.

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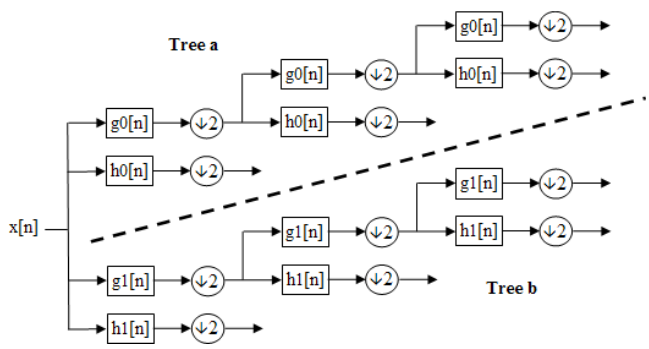

Fig. 5. Block diagram for a 3-level DTCWT

It acquires properties of directional selectivity and near shift invariance not achievable with the critically sampled DWT. These properties result in increased performance in signal analysis. The DT-CWT was invented by Kingsbury in 1998. It is shift invariant and allows directional wavelets in two higher dimensions. The dual-tree transforms demonstrate how complicated wavelets with desirable characters can be created, as well as a variety of signal and image processing applications.

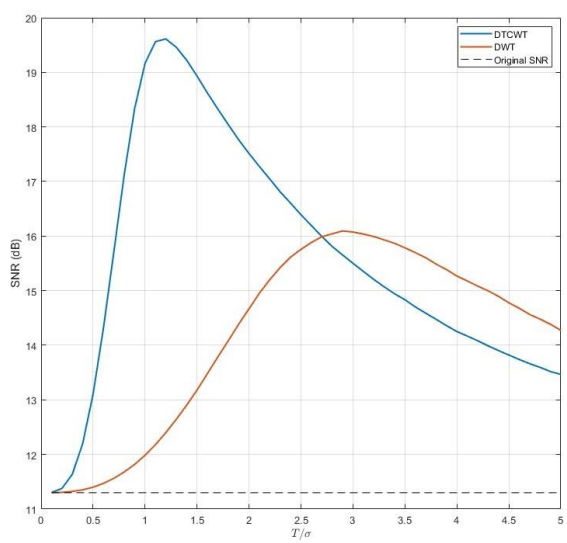

Fig. 6. SNR comparsion of DWT and DT-CWT

C. Gray Level Co-occurrence Matrix(GLCM)

The statistical texture analysis is utilised to derive texture features. Statistics are divided into first-order, second-order,and higher-order . Second-order statistical texture features were extracted by using Gray Scale Co-occurrence Matrix(GLCM). A number of applications are following this approach.

Properties of GLCM:

- The GLCM is of square shape.
- The quantization levels are equal the number of rows and columns.
- It is symmetrical about diagonal.

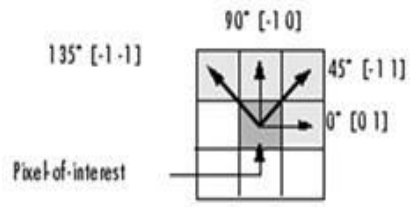

Fig. 7. Directions of GLCM

Steps involved in GLCM calculations:

1) Create a NxN GLC matrix, with N equaling the number of grey levels. The matrix is constructed as follows:
   - Let s be a pixel of interest of the image.
   - Assume W is a set of samples that surrounds s.
   - Using samples from the set W, determine each GLCM element i,j as the numerous times two samples with levels of intensity I and j occur in a defined spatial relationship.
2) Check the GLCM is symmetric or not. If GLCM is not symmetric then add a transposed copy of GLCM to itself.
3) Divide each GLCM element by the sum of all GLCM elements to normalize the GLCM.
4) Extract the Texture features from GLCM. The features extracted from normalized symmetric GLCM are:
   - Energy
   - Entropy
   - Contrast
   - Homogeneity
   - Variance

Energy: The texture uniformity, which is measured in pixel pair repetitions, is measured by energy.

$$Energy = \sum_i \sum_j p_{ij}^2 \tag{1}$$

Entropy: Entropy identifies the complexity of an image or disorder

$$Entropy = -\sum_i \sum_j p_{ij} log_2 p_{ij} \tag{2}$$

Contrast: Contrast identifies the difference between the lowest and highest pixel intensity in the image.

$$Contrast = \sum_i \sum_j (i-j)^2 p_{ij} \tag{3}$$

Homogeneity: Homogeneity measures the closeness sharing of components in the GLCM to diagonal.

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$$Homogeneity = \sum_i \sum_j \frac{1}{1+(i-j)^2} p_{ij} \quad (4)$$

Variance: This is a statistical measure of heterogeneity that is highly correlated with first-order statistical variables like standard deviation.

$$Variance = \sum_i \sum_j (i-\mu)^2 p_{ij} \quad (5)$$

where μ is mean of p_{ij}.

D. Neural Networks

1) Probabilistic Neural Network(PNN): In pattern recognition and classification issues, a probabilistic neural network offers a scalable alternative to back propagation neural networks. It doesn't require the huge backward and forward calculations that regular neural networks require.

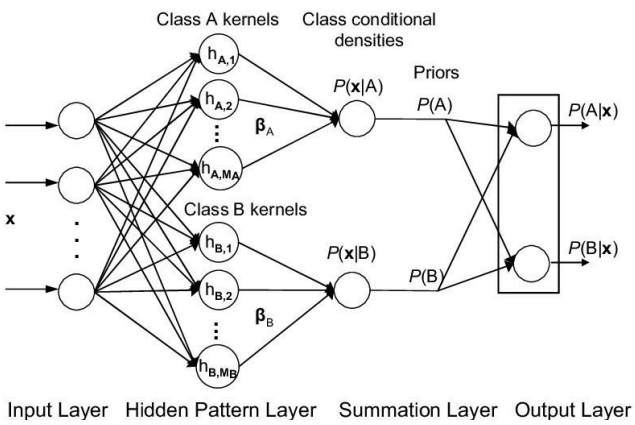

Fig. 8. Architecture of PNN

A non-parametric function and A Parzen window are employed in PNN technique to approximate each class's PDF. Probability distribution function of every class is used to estimate class probability of incoming input data. The Baye's rule is then used to given input data to the class along with highest posterior probability. loss are lowered by using this strategy. The Kernal Fisher discriminant analysis and Bayesian network are used to create PNN (statistical algorithm). The four tiers in PNN carry out the procedures.

They are as follows:

1) Input Layer
2) Pattern Layer
3) Summation Layer
4) Output Layer

For each o class, each sub-network of the PNN is just a Parzen window pdf estimator. The set of measurements is represented by the input layer nodes. The Gaussian functions produced by employing the specified number of data points as centres make up the pattern layer. For each class, the summation layer executes averaging operations on the outputs from the pattern layer. To determine which result is the most meaningful, the Output layer conducts a vote. Decision criteria are then used to determine the corresponding class label.

2) Convolutional Neural Network(CNN): Convolutional Neural Networks (CNNs) are same as traditional neural networks, except in neural networks the input is a vector, the input in CNNs is a multi-channeled image. This algorithm does not classify the tumour using the features extracted in the feature extraction step. To classify the tumour, it employs a neural network generated from segmented images of training data. Depending on the amount of training data, training can take minutes or even hours. Convolutional Neural Networks (CNNs) are neurons with learnable weights and biases that are similar to traditional neural network model. Each neuron takes many inputs, computes a weighted total of them, send them through activation function, and then returns output.

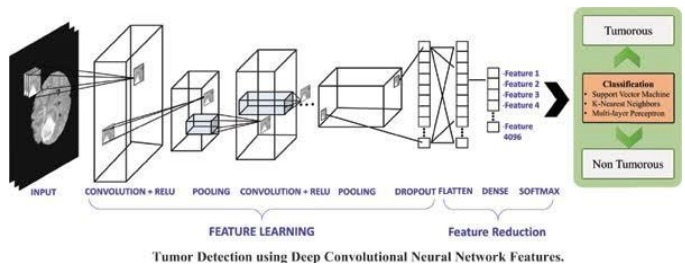

Fig. 9. Architecture of CNN

It has following layers: input layer, convolution layer, pooling layer, and output layer. The value of every neuron parameter is set to the same parameter, such as weight sharing, i.e.to deconvolution an image, a neuron with corresponding convolution kernels is used. Convolution and sampling are the two main processes in the CNN algorithm. The convolution process to deconvolute the input image a trainable filter Fx is used, and after that it adds a bias bx, resulting in the convolution layer. The sampling procedure involves pooling 'n' number of pixels from each of its surroundings to form a pixel, after that weighting by scalar weighting, adding the bias, and finally the activation function produces a narrow 'n' time feature map. CNN is advantageous since it explicitly avoids feature extraction and attempts to learn implicitly from the training dataset.

E. Spatial Fuzzy C-Means(SFCM)

The Spatial Fuzzy C-Means clustering algorithm incorporates spatial information, and membership weighting of each cluster is changed once cluster placement in the surrounding area is considered. First stage is same as the standard version. In the spectrum domain, FCM is used to compute the membership function. The membership

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information of second pass is: The spatial domain and spatial domain are mapped to each pixel. From there, a function is created. Iteration of the FCM continues because of membership that has been integrated with the function in space iteration will finish when maximum number of iterations has been reached. The variation between membership functions at two iterations should be less than the threshold value. J. C. Anderson first proposed the FCM method. Bezdek is a word that comes to mind when people think of the word FCM. FCM is based on the concept of applying weights to reduce total weighted mean-square error:

$$J(W_{qk}, z^k) = \sum_{k=1,K} \sum_{k=1,K} (W_{qk})||x^q - z^k||^2 \qquad (6)$$

$$\sum_{k=1,K} (W_{qk}) = 1 \qquad (7)$$

The SFCM algorithm combines the image's spatial information and reduces the traditional FCM algorithm's noise sensitivity. It is used to extract the image's edge information.

VII. RESULTS AND DISCUSSIONS

The proposed model simulation results are shown here. The algorithms were created in Matlab 2014b and operated on a PC with an Intel Core i3 processor, 4GB RAM, 64-bit Windows 10 operating system. The method was examined using MRI scanned pictures of the brain which are affected by noise. The proposed algorithm's performance was verified by eliminating those noise and represents the stage of tumor, its location and area effected by the tumor.

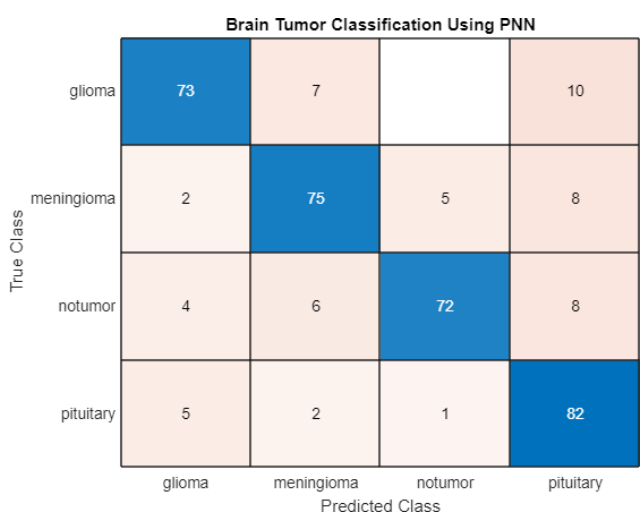

Fig. 10. Confusion Matrix of PNN

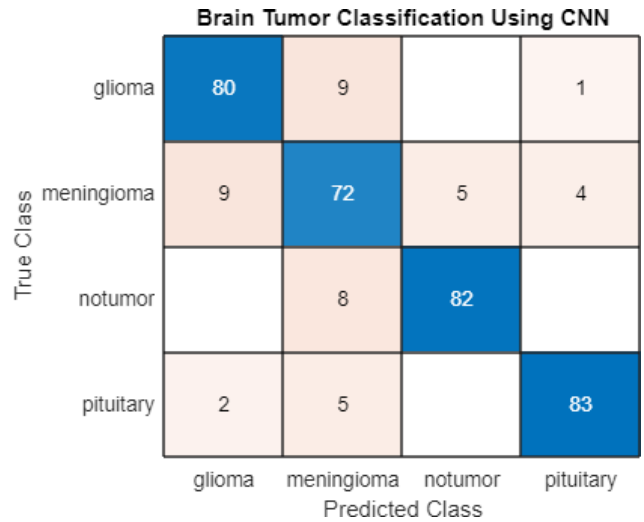

Fig. 11. Confusion Matrix of CNN

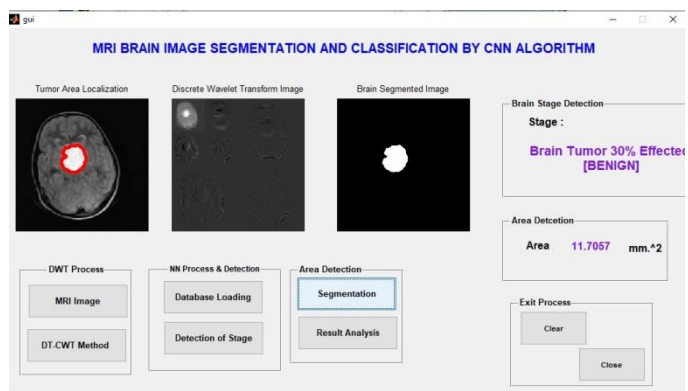

Fig. 12. Simulation Output

Parameters	PNN	CNN
Accuracy	83.6	88.3

TABLE I
COMPARISON OF CNN AND PNN ACCURACY.

VIII. CONCLUSION

This survey paper gives a brief overview of brain tumor segmentation and classification technique. A comprehensive analysis of various stages of image processing are used. We can conclude that algorithms and parameters in model are intended to improve system's efficiency in achieving better results. Features derived by using this algorithms improve effiency by extracting minute facts well about tumor . Reliability and accuracy are critical in tumour diagnosis .This methodology aids in enhancing accuracy and attaining the desired results.

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